Literature Review: Towards an Adaptive, Fully Automated Trading System by Integrating Self-Supervised, Meta and Reinforcement Learning

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Introduction

This literature review explores the novel juncture of financial trading and machine learning, in the context of developing an adaptive, fully automated trading system. The project explores the intersection of self-supervised learning, meta-learning, and reinforcement learning, with the aim of constructing a robust and generalisable trading agent that can adapt to different asset classes and market regimes. Financial markets are notoriously volatile, noisy, and non-stationary, making it exceedingly difficult for traditional rule-based systems. Recent academic and industrial studies have explored isolated components of such systems, including deep neural networks for time series forecasting, reinforcement learning for portfolio optimization, and, more recently, meta-learning for cross-domain adaptation. Methodologically, this review draws on deep learning frameworks for time series classification, adaptive reinforcement learning, and López de Prado's financial machine learning principles, including triple-barrier labeling, walk-forward validation, and purged cross-validation methods. The goal of this review is to evaluate existing work in these areas, identify gaps in holistic solutions, and position the planned system in existing best practices and research horizons.

1 Literature Search Process

For this review, the search was performed systematically using academic databases like Google Scholar, IEEE Xplore, and arXiv, which were initially consulted using selected search keywords and phrases. These included hybrids such as "self-supervised learning for financial time series," "deep learning for time series prediction," "reinforcement learning finance," "meta-learning adaptation," and "self-adaptation models." Boolean operators (OR, AND) were used effectively to restrict results and enhance specificity, particularly with combinations such as "contrastive OR generative self-supervised learning," "reinforcement learning AND portfolio optimization," and "meta-learning AND finance."

Peer-reviewed journal papers, high-impact conferences, and well-established preprint servers in the last five years were placed first to give a snapshot of the best current practices. Relevance was determined on an exact match basis with primary themes of the study: methodological quality of studies, empirical evidence in financial contexts, and findings on adaptive or generalisable modeling. Following abstract screening and critical appraisal, some key references were selected, such as Zheng et al. (2017) on deep neural architectures, MIT's SEAL framework (MIT, 2023) for meta-adaptive reinforcement learning, and Liu et al. (2024) on self-supervised learning paradigms for time series.

In addition, relevant grey literature and industry implementation were identified and examined through GitHub repositories and financial technology journals to present practical benchmarking against solutions at hand.

2 Academic Literature Review: Summary and Evaluation

2.1 Deep Learning in Time Series Data

One of the most significant deep learning efforts in time series classification (TSC) is Zheng et al.'s (2017) large-scale benchmark, which systematically compares the performance of

multiple distinct deep neural network (DNN) architectures across 97 diverse datasets of univariate and multivariate time series, comprising 8,700 runs combined (1; 2). This work, along with other studies of the time, initiated a methodological renaissance in TSC by demonstrating that discriminative end-to-end deep models not only outperform traditional statistical and feature-engineered approaches but are also more generalizable and scalable (3; 2).

Among the architectures tested, Residual Networks (ResNet) and Fully Convolutional Networks (FCN) stood out as top performers (1; 3). ResNet's popularity stems from its innovative use of skip connections, which enable direct gradient propagation throughout the network and help evade the vanishing gradient problem specific to deep architectures. Such architecture enables ResNets to learn richer, longer-range temporal dependencies. This preference is particularly beneficial for financial time series, where lagged effects and intertemporal correlations (e.g., regime switching, market momentum) play a significant role. In contrast, FCNs dispense with fully connected layers in favor of a lightweight convolutional structure with global average pooling (GAP), which reduces parameter complexity, enhances computational efficiency, and promotes generalizability, especially in limited data or real-time conditions (3). FCN's design is especially suited for financial systems that must run at scale or within latency constraints.

One significant advancement over Zheng et al. and subsequent reviews is the rigorous use of Class Activation Maps (CAM) for post-hoc model explainability (3; 2). By identifying which intervals or features of the input had the most significant contribution to a classification decision, CAMs provide beneficial transparency, addressing regulatory and business requirements for explainability in high-risk domains, such as finance. This aims to bridge the gap between the predictive power of deep learning and the need for explanation and confidence in automated financial decision-making.

Recent works, for instance, Fawaz et al. (2019) and Foumani et al. (2024), extend even further to establish that discriminative, end-to-end deep models, particularly CNN-based and residual models, not only perform better but often outperform state-of-the-art performance on TSC tasks in various domains (2; 4). The reasons for this are that they can hierarchically extract multi-scale temporal features directly from the raw signal without requiring brittle domain-specific feature engineering. This confers strength in terms of non-stationarity and regimes of unobserved data—a central issue in financial markets—and also enables broader applicability and transferability.

Basing its approach on these fundamental findings, the proposed system utilizes a CNN-based encoder design, culling design motifs from ResNet and FCN, to maximize both explanation and accuracy of feature representations extracted from unlabeled price data. Features learned here are further tuned by downstream reinforcement learning and meta-learning modules to construct an effective and tightly explainable pipeline. To further secure trust and accountability, the plan is to integrate CAM-based interpretability or similar attention visualisation techniques into our models, so that model decision-making will always be transparent and auditable according to best practice and regulatory expectations.

2.2 Meta-Learning for Adaptive Agents

The most critical challenge is the domain shift, the fact that learning patterns from one asset or state of the market do not necessarily apply to others. Supervised and reinforcement learning models assume that the data distribution is stable; however, this

assumption is often violated in real financial systems, where asset behaviours, volatility regimes, and liquidity profiles continuously shift. This limitation gives rise to the interest in meta-learning, offering a basis for rapid model adaptation across a broad variety of tasks or domains.

Meta-learning algorithms are designed to learn a model on a task distribution so that it rapidly generalizes to novel, unseen tasks based on small quantities of new data. Financially, this equates to a system that, having been trained on a fixed set of assets (e.g., equities, cryptocurrencies, foreign exchange), can react to a novel asset with minimal retraining—a capability that is crucial in volatile markets. One of the most well-known methods in this domain is Model-Agnostic Meta-Learning (MAML), which acquires an adapted initialisation of parameters that may be quickly tuned through a series of gradient steps on new tasks (5).

While direct applications of MAML and meta-learning to trading are still in their nascent stages, MIT's SEAL (Self-Edit for Adaptation in Language Models) is a conceptual innovation. SEAL introduces a novel form of self-adaptation in large language models (LLMs) where a model trains to generate and execute edits on its parameters when it makes incorrect predictions (6). This edit process is regulated by a reinforcement learning loop, with the downstream performance of the edited model as a reward signal. The most significant innovation is the edit encoder that proposes structured changes (parameter deltas) and enables the model to improve without complete retraining.

While SEAL is specifically for use in NLP and assumes access to real-time, supervised feedback (e.g., correct text generation), its architecture-agnostic nature and reinforcement-based adaptation methodology are highly relevant to financial systems. However, as the initial study and subsequent analysis noted, SEAL is highly limited in finance. First, it requires access to ground-truth labels shortly after prediction, which is not the case in trading, as outcome labels (e.g., whether a trade was profitable or not) are available only with some delay. Second, SEAL suffers from catastrophic forgetting, where repeated updates degrade performance on previous tasks, so that it is not suitable for time-evolving tasks unless properly controlled.

As opposed to direct use of SEAL, our method will aim to leverage it's reinforcement-based meta-update ideology to some extent. However, it substitutes edit encoders with a more consistent, interpretable, and better-comprehended meta-learning framework — MAML. MAML does not need task-specific generation of edits; instead, it optimizes for a generalizable initialization that can be fine-tuned rapidly with a handful of gradient steps on new tasks or assets (5). This is more naturally suited to financial applications, where learning on new equipment must be done without being retaught from scratch, but not at the expense of destabilising existing knowledge.

In practice, the system in question will use MAML to pretrain on various asset classes so that the base model learns a general representation of market conditions. When deployed, it can effectively adapt to a target asset (such as a new listing stock or an emerging-market currency pair) based on very little historical data. This meta-adaptive pretraining layer is positioned between the reinforcement learning agent and the self-supervised encoder to ensure that representations are flexible and pertinent as regimes shift in finances.

2.3 Reinforcement Learning for Portfolio Optimization

Reinforcement learning (RL) has gained significant traction in the finance domain, particularly for uncertainty-based sequential decision-making tasks such as portfolio man-

agement, asset allocation, and high-frequency trading implementation. In an RL system, an agent deals with an environment (for instance, a financial market), making decisions (actions) based on observed data (states) in order to maximize cumulative rewards (for example, risk-adjusted returns). This setup is technically represented as an MDP, with the policy of the agent learning to map states (e.g., price trend, volatility, technicals) to actions (buy/sell/hold) to optimize a long-term performance objective.

Some of the key contributions here, for instance, by Jiang et al. (2017), demonstrate how the use of RL algorithms, e.g., Deep Q-Networks (DQN), Deep Deterministic Policy Gradient (DDPG), and Proximal Policy Optimization (PPO), can be effectively utilized to control portfolios (7). These algorithms differ in the way they optimize policy: DQN is value-based and well-suited for discrete action spaces, whereas DDPG and PPO are actor-critic algorithms that are better suited to continuous action spaces and, therefore, better suited to problems such as portfolio rebalancing and trade sizing (7; 8). PPO in particular has gained wider usage because of its relative stability and sample efficiency in high-dimensional stochastic worlds.

One of the greatest strengths of RL in financial use is that it has the capability to learn to respond to feedback over time and shift strategies dynamically according to evolving market conditions. Unlike static machine learning algorithms or rule-based systems, RL agents learn from their personal trading history, frequently refining their policy utilizing the reward feedback they receive. This adaptive quality makes RL an appealing framework for real-time, self-adjusting trading systems.

But there are a number of important limitations also described in the literature. RL models tend to require large amounts of data along with enormous exploration, which is problematic given the limited availability of good-quality labeled financial data. They are also sensitive to reward function design — poorly designed rewards can lead to undesirable behaviour (e.g., over-trading or risk-taking). Furthermore, there is a gap between reality and simulation: learned policies in back test settings are not generalizable to actual markets since assumptions are not realistic as market impact is not simulated (7). There is the risk of overfitting compounded by information leakage whenever upcoming data inadvertently influences training using flawed validation methods.

To achieve this, current studies increasingly emphasize employing statistically sound validation methods. Techniques such as triple-barrier labeling allow for more realistic definition of trades by defining entry/exit points in terms of prices and temporal windows (9). Purged k-fold cross-validation is employed to prevent data overlap between the training and test sets in the case of time series. Walk-forward validation simulates real-world deployment through the use of past data to train and unknown future data to test. These are the best practices to reduce lookahead bias and ensure robust out-of-sample testing, which is critical to the validity of RL-based financial models.

Informed by these findings, our proposed system employs PPO for updating policy stability and suitability for ongoing trading activity. It further involves the application of walk-forward testing, purified cross-validation, and triple-barrier-based labelling in training and testing processes such that performance measures are kept realistic to deployment settings. These methodological choices put the system under the current best practices of reinforcement learning in finance without suffering from known flaws with good design and validation.

3 Review of Existing and Related Projects

3.1 Financial reinforcement learning (FinRL)

FinRL is an open-source, modular library developed by the AI4Finance Foundation for training and deploying DRL agents such as DDPG, PPO, and A2C for trading stocks. It allows real-time execution by using broker APIs like Alpaca and offers data preprocessing, feature engineering, and backtesting for historical data. FinRL agents rely heavily on traditional technical indicators (MACD, RSI, Bollinger bands) as input features, and implement task-specific training protocols, in which a model is trained and tested for a particular asset or market regime (10). While FinRL demonstrates the effectiveness of using RL to make financial decisions, its architecture has some drawbacks. The use of pre-fixed technical indicators restricts the learning of the model from source market data and can cause the model to generalise inadequately between assets. Additionally, the lack of meta-learning mechanisms means that the system must be re-trained from scratch when used on new instruments or market environments. Finally, its backtesting framework does not incorporate statistically robust verification techniques such as triple-barrier tagging, purged k-fold cross-validation, or walk-forward validation, leaving it vulnerable to information spillover and overfitting (9).

3.2 FinBERT: Financial Sentiment Analysis

FinBERT, a financial text-tuned BERT model, has been widely used for sentiment analysis in finance (11). FinBERT is typically implemented in trading systems whose rule-based actions are prompted by sentiment output—i.e., trading when news sentiment is classified as positive. FinBERT has been shown to outperform traditional bag-of-words models in classifying financial news with richer contextual information into nuanced language in earnings reports, analyst commentary, and market commentary.

Although these are positives, FinBERT systems do possess some drawbacks. They are typically static classifiers, producing sentiment scores without learning or changing over time. Decision logic is likely to be hard-coded, with no feedback channel for policy revision based on real-time outcomes. Additionally, FinBERT systems must have access to labeled textual data, something that is not always available, especially in thin markets or on new issuance. Their lack of direct ability to ingest market data also limits their generalisability and responsiveness.

Conclusion

This review has compared, critically through systematic searching and selection, three foundation studies: advanced financial time series classification architectures (Zheng et al., 2017), state-of-the-art reinforcement-driven self-adaptive strategies (MIT SEAL, 2025), and comparative evaluations of self-supervised learning approaches to sequential data (Liu et al., 2024). Real-world practices, including modular software such as FinRL and FinBERT-based sentiment trading platforms, were also used as baselines, indicating the current state of practice. Key findings are that, with considerable improvements in each of the subdomains, most of the systems implemented remain in silos and are hampered by rigid architectures or lack of testing against real-world market regimes. Deep learning models, ResNet and FCN, are state-of-the-art when used to work on time

series data, best in terms of accuracy and feature discovery; yet they still fail to perform transferability across asset classes and regimes of markets. Reinforcement learning methods, despite being extremely successful in capturing sequential decision making, tend to be reward-sensitive in design and subject to overfitting and narrow in competence in the actual environment. The promise of meta-learning and self-supervised learning, a broad generalisation that allows for fast adaptation with less supervision, remains unrealised mainly in production trading systems. However, research has investigated their contribution to model robustness and adaptability.

These have encouraged the idea behind the research project, which seeks to address these gaps by proposing an end-to-end trading architecture that integrates self-supervised representation learning, model-agnostic meta-learning, and state-of-the-art reinforcement learning agents. This combined solution is distinguished by the use of robust validation protocols adopted from best academic and industry practice, including triple-barrier labelling, purged cross-validation, and walk-forward testing, to ensure that performance metrics reflect genuine out-of-sample reliability.

From an industry and research perspective, the proposed research project addresses the primary needs for scalability, versatility, and transparency in financial automation frameworks. The objective is to provide a practical and versatile solution tailored to the complexities and constant transformations present in current trading environments.

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